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## **Abstract**

Ransomware is now one of the most serious and fast developing cyber security threats, as it is able to encrypt data and negatively affect the critical operations of individuals, businesses and systems of countries. The study examines ransomware activity in a host environment, including the communication between ransomware and the file system and the discussion of the effective methods of preventing and mitigating it. This study addresses the following key research question: How does ransomware manipulate a host's file system during an attack, and what proactive strategies can be implemented to prevent or reduce its effect?

This research employed a multi-phase methodological framework incorporating both behavioral examination and preventive assessment. During the behavioral stage, the controlled experiment was performed under closed virtual environments to trace ransomware activity in real time. System level changes including unauthorized file encryption, privilege escalation and system directory modification were recorded by use of analytical tools like process monitors, registry trackers, and file integrity verification tools. During the preventative stage, the different defense measures, including heuristic scanning, anomaly detection, and access control policies, were studied to determine their efficiency in curbing the recent types of ransomware.

The results demonstrate that ransomware attacks are characterized by the series of the same behavior that is usually based on the infiltration of the system, file enumeration, then encrypted and deleted recovery files including shadow copies. The researchers noted that ransomware changes file extensions, modifies registry keys in order to be persistent, and uses either symmetric or hybrid encryption algorithms to encrypt user data. Behavioral monitoring indicated some unique trends in accessing files and creating processes, which can be used as precursors to an attack. Relational analysis shows that behavioural based and machine-learning based detection systems are more effective than the traditional signature-based antivirus techniques, especially when dealing with polymorphic ransomware that continuously changes its code in order to go undetected.

The study finds that the most effective method of mitigating the risk of ransomware is a layered defense response. This involves the adoption of real time monitoring of file systems, the application of least privilege principles, regular offline file system backups, and incorporation of smart file system anomaly detection models. Understanding file system behavior of ransomware will be able to give cyber security professionals a basis on which they can devise proactive defense measures that can detect and prevent ransomware before it can encrypt the files.

The study is relevant to the evolving body of knowledge on cyber security defense in that it provides empirical evidence about the technical behavior of ransomware and demonstrates viable prevention methods that can be utilized in various computing environments.

## 1. Introduction

Ransomware has evolved from a simple nuisance into one of the most disruptive threats targeting modern computing hosts. What makes it uniquely dangerous is its habit of going straight for the heart of a system: the file system. Once ransomware lands on a machine, it wastes no time in crawling through directories, encrypting anything that looks even remotely important, and making sure victims have very few recovery options left, unless they are willing to pay for their own data, of course. As host systems continue to integrate cloud services, local storage, and hybrid execution models, the opportunities for attackers have expanded just as quickly as the defenses built to stop them.

This research takes a practical and somewhat hands-on view of ransomware by analyzing how it interacts with the file system on a host and how we can detect these behaviors early. To keep things grounded, we study both documented real world ransomware behavior (most notably *WannaCry*, the infamous star of many security textbooks) and the behavior of an educational malware tool called *Unick Daemon*, developed by one of our team members. While *Unick Daemon* is not ransomware, it shares some suspiciously similar behaviors. i.e. persistence, file-system modification, and remote arbitrary code execution which makes it a perfect comparison point for understanding how easily normal malware can drift into ransomware territory if someone with the wrong intentions adds a few extra lines of code.

The overall goal is simple: understand what ransomware does, watch how these behaviors appear at the filesystem level, and identify what defensive strategies can detect and stop the attack before the user has to Google “how to recover encrypted files without paying ransomware.” Through this analysis, we highlight why host level monitoring and layered defenses remain essential for protecting everyday systems from increasingly sneaky attacks.

## 2. Background and Related Work

Ransomware attacks typically follow a predictable lifecycle: infiltration, execution, enumeration, encryption, cleanup, and finally extortion. Even though different families implement this process in their own artistic ways, the file-system portion is surprisingly consistent across strains. Earlier families such as *CryptoLocker*, *CryptoWall*, and *TeslaCrypt* set the stage for modern ransomware by using symmetric encryption and spreading through email attachments. Modern examples like *WannaCry*, *Ryuk*, *LockBitm*, take this to an entirely new level with worm-like spreading, kernel manipulation, and hybrid encryption models.

## **2.1 File System Interaction in Ransomware**

The central element of any ransomware attack is the file system. Research shows that nearly every strain begins by enumerating directories recursively, picking out common file extensions (.docx, .pdf, .jpg), then encrypting these files with AES or a similar algorithm. Studies such as Scaife et al. (2016) show that ransomware-generated disk I/O patterns are unusually bursty and sequential, making them observable even with lightweight monitoring systems. Other works highlight that ransomware frequently tampers with OS-level file recovery mechanisms, most notably deleting shadow copies on Windows.

## **2.2 Detection Techniques**

Historically, signature based detection has been the go-to approach for stopping ransomware, but it fails miserably against modern variants that constantly mutate. Behavioral analysis became the next logical evolution. Systems like HDD based anomaly detectors, host based intrusion detection systems (HIDS), and machine learning powered EDRs analyze unusual file modifications, process spawning, and access to sensitive directories. Researchers have shown that behaviors such as mass file renaming, rapid encryption bursts, and abnormal registry modification are strong indicators of active ransomware.

## **2.3 Persistence and File-System Manipulation in Malware**

Interestingly, many non-ransomware tools, especially Remote Access Trojans (RATs), exhibit behaviors that look deceptively similar to ransomware's early-stage behavior. RATs typically maintain persistence, manipulate files, and run arbitrary commands, all of which can mimic the initial phases of a ransomware attack. This creates opportunities for attackers to modify existing tools into ransomware with minor adjustments, an idea that becomes important in our analysis of Unick Daemon later in this paper.

## **3. Methodology / Technical Framework**

To understand how ransomware interacts with a host's file system and how it can be detected early, we used a two stage research approach combining both documented ransomware behavior and a hands-on lab experiment. The first stage focused on analyzing existing research and forensic reports of the WannaCry ransomware family. The second stage involved executing and monitoring Unick Daemon, a Python-based remote access tool (RAT) created by one of the team members, inside a controlled Windows 10 virtual machine. While not ransomware, Unick Daemon shares structural elements. i.e. persistence, file modification capabilities, and arbitrary code execution that allow it to mimic early ransomware behaviors when tweaked.

### 3.1 Environment Setup

All practical experimentation occurred within a Windows 10 virtual machine to ensure isolation, repeatability, and safety. The VM was kept fully offline to prevent unintended connections or malware spread. For monitoring, we used:

1. **Process Monitor (ProcMon)**: to log filesystem activity, registry changes, process creation, and thread behavior.
2. **Wireshark**: to observe reverse connection patterns, especially during Unick Daemon's communication attempts.

These tools allowed us to track both host level activity (ProcMon) and network behavior (Wireshark), without overwhelming the analysis with too many utilities.

### 3.2 WannaCry Reference Analysis

Instead of executing WannaCry (for obvious reasons), we relied on forensic analyses, incident reports, and peer reviewed literature. This included sources from academic papers, Microsoft security bulletins, CERT analyses, and malware breakdowns from reputable cyber security labs.

Key behaviors extracted from these sources included:

- Directory crawling through common user paths.
- Bulk file encryption using AES.
- Appending its trademark .WNCRY extension.
- Deleting Volume Shadow Copies using *vssadmin*
- Creating persistence via registry keys.
- Initiating a killswitch domain lookup.
- These behaviors formed the “baseline ransomware profile” used in comparisons.

### 3.3 Unick Daemon Experiment

Unick Daemon, built as an educational RAT, was executed inside the VM and allowed to connect to our command server (Just another PC with admin program) using its reverse connection mechanism. Its notable capabilities included:

- Remote arbitrary Python code execution
- File upload/download
- Persistence through startup folder placement and self-monitoring
- Renaming and regenerating its executable to evade manual removal attempts
- Automatic hotspot based reconnection when available

ProcMon logs were collected while the RAT initialized, connected, self-replicated, renamed files, and received remote commands. These behaviors were later mapped against early stage ransomware actions to examine overlap.

### 3.4 Comparative Framework

To maintain consistency, we categorized observed activity into four behavioral groups:

- File-System Enumeration & Access Patterns
- Modification, Encryption, or Alteration of Data
- Persistence and Survivability Behavior
- Remote Execution and Command Control Flow

This four part model allowed us to compare how legitimate ransomware families (WannaCry) and adaptable RAT tools (Unick Daemon) use the file system in similar or different ways. The comparison also highlights how easily malware can evolve into ransomware with minimal changes to code.

## 4. Case Study / Experiment Results

This section summarizes the behavioral findings from both the WannaCry analysis (literature-based) and the Unick Daemon experiment (hands-on). While they differ in intent and complexity, both tools reveal interesting overlaps at the file-system level.

### 4.1 Documented Behavior of WannaCry

Based on multiple forensic analyses, WannaCry exhibits a textbook ransomware execution pattern:

#### 1. File Enumeration

WannaCry aggressively scans directories, beginning with:

- a. Desktop
- b. Documents
- c. Downloads
- d. Network shares

It identifies files with commonly used extensions such as .docx, .pdf, .xlsx, .jpg, and .mp3, preparing them for encryption.

#### 2. File Encryption & Extension Changes

The encryption workflow includes:

- a. Generating a unique AES key
- b. Encrypting files sequentially
- c. Renaming them with the .WNCRY extension
- d. Dropping a ransom note in each affected directory.

The sequential write burst pattern is highly irregular and easily detected in ProcMon-like logs.

### 3. Shadow Copy Deletion

WannaCry removes Windows shadow copies using the command *vssadmin.exe Delete Shadows /All /Quiet*

This prevents users from restoring earlier file versions.

### 4. Persistence

It modifies registry paths under:

- a. *HKLM\Software\Microsoft\Windows\CurrentVersion\Run*
- b. *HKCU\Software\Microsoft\Windows\CurrentVersion\Run*

These entries force the ransomware to relaunch automatically.

### 5. Cleanup & Ransom Delivery

The malware wipes temporary files, drops @Please\_Read\_Me@.txt ransom notes, and displays its infamous GUI demanding payment via Bitcoin.

## 4.2 Unick Daemon Hands-on Behavioral Experiment

Now for the fun part: the behavior of our own RAT inside the VM.

### 1. Startup and Reverse Connection

Now upon execution:

- a. Unick Daemon copies itself to the Windows startup directory.
- b. It also writes a copy into the root of C:\.
- c. It tests for the presence of a specific Wi-Fi hotspot name frequently.
- d. If detected, it initiates an automatic reverse connection to the operator.

Wireshark captured this connection pattern, showing consistent outbound traffic attempts to the operator's IP/port, confirming its NAT bypassing design.

### 2. Persistence and Self-Monitoring

ProcMon logs showed repeated attempts by the RAT to:

- a. Check whether the startup folder copy still exists.
- b. Regenerate or rename itself if modified.
- c. Monitor its own directory for tampering.

This behavior mimics ransomware's attempts to survive reboots or user cleanup efforts.

### 3. Arbitrary Code Execution

Unick Daemon's most powerful capability was the ability to execute remote Python code along with:

- a. Listing directory structures
- b. Uploading or downloading files
- c. Running system-level commands
- d. Writing or deleting files

Although the RAT did not encrypt files, the presence of remote execution means that adding file-encryption logic would be trivial.

#### 4. Potential for Ransomware-Like Behavior

Because of its persistent foothold, direct filesystem manipulation, and arbitrary code execution ability, Unick Daemon could realistically be transformed into ransomware with surprisingly little modification. For example, a simple AES encryption module plus a directory walker function could redefine its purpose entirely.

This makes it an excellent contrast tool for understanding filesystem behavior differences between “ordinary” malware and full ransomware.

#### 4.3 Comparison Table: WannaCry vs Unick Daemon

| Behavior Category     | WannaCry        | Unick Daemon                  | Overlap |
|-----------------------|-----------------|-------------------------------|---------|
| Directory Enumeration | Aggressive      | On demand (via remote code)   | Medium  |
| File Encryption       | Core function   | Not implemented               | Low     |
| File Renaming         | Adds .WNCRY     | Self-renaming for persistence | Medium  |
| Shadow Copy Deletion  | Yes             | No                            | Low     |
| Persistence           | Registry        | Startup folder + self-check   | High    |
| Arbitrary Code Exec   | No              | Yes                           | Medium  |
| Network Behavior      | Spreads via SMB | Reverse connection            | Medium  |
| Ransom Delivery       | GUI ransom note | None                          | None    |

Table 1. Comparison between WannaCry and Unick Daemon

### 5. Results and Discussion

The combined analysis of documented ransomware behavior and the experimental observations from Unick Daemon produced several interesting insights into how malware interacts with the host file system. While WannaCry represents fully weaponized ransomware with a well defined encryption workflow, Unick Daemon, despite not being ransomware, still demonstrated behaviors that overlap significantly with early ransomware stages. This overlap highlights why host level monitoring is so important and why even “ordinary malware” should be treated as a potential precursor to a more dangerous attack.

#### 5.1 Detectable File-System Patterns

One of the clearest findings is that ransomware produces distinctive filesystem patterns that are difficult to hide. WannaCry, for example, generates massive bursts of file reads and writes, consistent renaming operations, and deletes recovery mechanisms such as shadow copies.

These patterns are quite different from normal user activity or even typical software updates. Behavioral research agrees that such patterns are detectable through lightweight filesystem monitoring, even before encryption completes.

Unick Daemon, on the other hand, showed a more subtle footprint but still interacted frequently with the file system, especially when establishing persistence or executing remote commands.

ProcMon logs displayed periodic checks on startup folder entries, file copy operations, and renaming loops. While not as noisy as ransomware, the persistence logic alone would stand out in a host based monitoring system.

## 5.2 Overlapping Behaviors That Signal Possible Ransomware

The most interesting overlap was the observation that many actions associated with ransomware are not unique to ransomware. Malware such as Unick Daemon can:

- recursively list directories,
- upload files elsewhere,
- modify file content,
- maintain persistence,
- escalate privileges (indirectly),
- log screen, mouse, keyboard.
- remotely execute arbitrary code.

This means the early phase of ransomware (file enumeration, directory scanning, privilege acquisition) can look nearly identical to the behavior of a flexible RAT. In other words, *“By the time ransomware specific behavior appears, it may already be too late to intervene.”*

This finding supports the growing consensus in host security research: detecting earlystage behavior is just as important as detecting encryption activity.

## 5.3 Why File System Monitoring Helps

A key takeaway is that hybrid monitoring, watching file I/O patterns, registry changes, and unexpected directory traversal, can catch both ransomware and “pre-ransomware-like” behaviors. For example:

- Continuous attempts to recreate a deleted startup entry.
- Unexpected renaming of executables.
- Sudden spikes in file access rates.
- Attempts to hide processes or mask activity.

These signals should raise alarms even in the absence of encryption.

## 5.4 Signature vs Behavioral Detection

Traditional signature-based detection fails against malware like WannaCry due to polymorphism, packers, code mutation, newly released versions.

Behavioral detection, especially when combined with machine learning approaches, offers better coverage by focusing on what the malware does, not how it looks. Our experiment reinforced this point: Unick Daemon could easily evolve into ransomware without changing its core architecture, meaning signatures wouldn’t catch the transition.

## **6. Limitations**

No study is perfect, and ours has several limitations worth noting:

### **6.1 VM-Based Testing**

All hands-on experiments were conducted inside a virtual machine. While this ensured safety, it may not perfectly represent real world host environments. Some malware behaves differently when it detects a VM, although Unick Daemon did not show such behavior.

### **6.2 Unick Daemon Is Not Ransomware**

Since Unick Daemon does not perform encryption or destructive operations, its behavior only serves as a partial comparison. It helps us understand pre-encryption stages but not the full ransomware lifecycle.

### **6.3 Limited Toolset**

Our analysis relied primarily on ProcMon and Wireshark. These tools provide great insight, but a more comprehensive toolset (e.g., Sysmon logs, registry diffing tools, EDR telemetry) would offer a deeper behavioral picture.

### **6.4 Controlled Network Conditions**

The RAT's reverse connection was only tested in predictable conditions (local hotspot and direct server). More advanced or evasive networking behavior was outside the scope of this study.

### **6.5 No Real Encryption Testing**

We did not integrate custom encryption modules into Unick Daemon for ethical and safety reasons. However, data are encrypted upon upload/download of a file. As a result, we could not test how our monitoring setup would behave under actual encryption workloads.

## **7. Conclusion and Future Work**

Ransomware continues to evolve rapidly, exploiting every layer of a host system. i.e. its file structure, registry, kernel, and even built-in recovery tools. By understanding its filesystem behavior, defenders can build smarter detection mechanisms that trigger alarms long before encryption finishes.

Through analysis of WannaCry and our experiment with Unick Daemon, we observed that:

- Ransomware is noisy when interacting with the file system.
- Many early ransomware behaviors overlap with generic malware.
- Persistence mechanisms and file monitoring routines are clear warning signs.
- Behavioral detection systems significantly outperform signature-based ones.

A layered defense, file integrity monitoring, least privilege access, behavioral EDR tools, and regular offline backups remains the most effective protection strategy.

## **Future Research Directions**

There are several ways this research could be extended:

1. Integrating Sysmon or EDR Telemetry
  - Collecting more granular system events would allow deeper behavioral modeling.
2. Emulating Ransomware Using a Safe Encryption Module
  - A sandboxed simulation of encryption would provide more realistic test data.
3. Testing Advanced Machine Learning Models
  - Training ML classifiers on filesystem events could reveal early warning signatures.
4. Studying Cross-Platform Ransomware Behavior
  - Expanding experiments to Linux and macOS hosts would broaden applicability.
5. Building a Lightweight Early-Warning Tool
  - A simple behavioral detector focused on file I/O anomalies could be an excellent follow-up project.

This study contributes to ongoing cyber security research by showing how detailed observation of filesystem behavior, even from non-ransomware malware, can reveal patterns that help detect attacks before the damage is done.

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